OPTIMISATION

1. a) Note that the function $\tilde{f}$ is quadratic in x and it is convex and radially unbounded, hence it has a unique global minimiser. This is obtained solving the equation

$$0 = \nabla \tilde{f}(x) = \nabla f(x_k) + \frac{1}{t}(x - x_k),$$

which yields the iteration

$$x_{k+1} = x_k - t\nabla f(x_k).$$

Note that this is the gradient iteration with t playing the role of the line search paramenter α_k. [4 marks]

b) The function f has a unique stationary point which is obtained solving the equation

$$0 = \frac{\partial f}{\partial x} = -2x + 2a,$$

yielding $x^\star(a) = a$ and $f(x^\star(a), a) = 5a^2$. Note now that

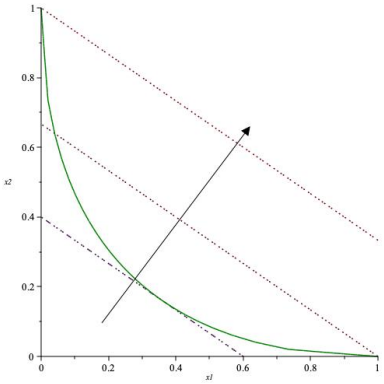
$$\frac{df(x^\star(a), a)}{da} = 10a$$

and that

$$\left.\frac{\partial f}{\partial a}\right|_{x=x^\star(a)} = \left.(2x + 8a)\right|_{x=x^\star(a)} = 10a,$$

which shows that the Envelop Theorem holds for this specific example. [4 marks]

c) i) The admissible set is the green curve in the figure, the level lines of the objective function are the dotted lines, and the arrow indicates the direction in which the objective function decreases.



From the graph we see that the solution of the problem is given by the point $P_m = (0,1)$, which gives a cost of $f = -3$. Note that the point $\tilde{P}_m = (1,0)$ is also a local minimiser, giving $f = -2$. Finally, the point $P_M = \left(\frac{9}{25}, \frac{4}{25}\right)$ is a global maximizer, giving $f = -6/5$. [4 marks]

ii) The Lagrangian for the problem is

$$L(x_1, x_2, \lambda) = -2x_1 - 3x_2 + \lambda(\sqrt{x_1} + \sqrt{x_2} - 1),$$

Answers

yielding the necessary conditions of optimality

$$0=\frac{\partial L}{\partial x_1}=-2+\frac{\lambda}{2\sqrt{x_1}},\quad 0=\frac{\partial L}{\partial x_2}=-3+\frac{\lambda}{2\sqrt{x_2}},\quad 0=\sqrt{x_1}+\sqrt{x_2}-1.$$

The point P_m (and also the point $\tilde{P}_m$) does not satisfy the necessary conditions since these are not defined at $x_1=0$ (and $x_2=0$). The reason why the conditions cannot be used is that the equation describing the constraint is not differentiable at P_m (and at $\tilde{P}_m$). Note, finally, that solving the necessary conditions yields the (global) maximiser P_M with optimal multiplier $\lambda=12/5$. [4 marks]

iii) Since $x_1\in[0,1]$ and $x_2\in[0,1]$ one could write

$$\sqrt{x_2}=1-\sqrt{x_1},$$

yielding $x_2=1-2\sqrt{x_1}+x_1$, hence

$$f_{red}=-5x_1+6\sqrt{x_1}-3,$$

which, in the interval $[0,1]$, has a global minimizer at $x_1=0$, a local minimizer at $x_1=1$ and a global maximizer at $x_1=9/25$, consistent with the results in parts i) and ii). [4 marks]

Answers

2. a) The Lagrangian of the problem is (note the change of sign of the objective function)

$$\mathscr{L}(K,L,\lambda) = -KL + \lambda(p_K K + p_L L - B).$$

The necessary conditions of optimality are

$$0 = -K + p_L\lambda, \qquad 0 = -L + p_K\lambda, \qquad p_K K + p_L L - B = 0.$$

These give the candidate optimal solution

$$K^\star = \frac{B}{2p_K}, \qquad L^\star = \frac{B}{2p_L}, \qquad \lambda^\star = \frac{B}{2p_K p_L}.$$

[4 marks]

b) The estimate gives an increase (note that we now consider the maximisation problem, hence an increase in budget gives an increase in production)

$$\delta f = \lambda^\star \delta B = \frac{1}{10}\ \frac{B^2}{2p_K p_L} = 0.05\frac{B^2}{p_K p_L}$$

The actual increase is obtained replacing the optimal solution obtained in part a) in the objective function, which gives $f^\star = \frac{1}{4}\frac{B^2}{p_K p_L}$, and *augmenting the budget*. This results in the new cost

$$f^\star_{new} = \frac{1}{4}\ (11/10)^2\frac{B^2}{p_K p_L} = 0.0525\frac{B^2}{p_K p_L},$$

hence

$$\delta^\star_f = f^\star_{new} - f^\star = \frac{1}{4}\ \left(\left(\frac{11}{10}\right)^2 - 1\right)\frac{B^2}{p_k p_L} = 0.0025\frac{B^2}{p_K p_L}.$$

Note that the estimate is reasonably accurate (within 5%). [4 marks]

c) Eliminating K gives the reduced objective function

$$f_{red}(L) = -\log\frac{B - Lp_L}{p_K} - \log L,$$

which has a stationary point at $L^\star = \dfrac{B}{2p_L}$.

By computing the second derivative one obtains that the function f_{red} is globally convex in its domain of definition, hence the stationary point $L^\star$ is a global minimiser for the log-transformed problem and hence of the original problem.

Note that the log-transformation does not alter the problem, since the logarithm is a strictly monotonic function. [6 marks]

d) The Lagrangian for this problem is

$$L(q_1, q_2, \cdots, q_n, \lambda) = -U(q_1, q_2, \cdots, q_n) + \lambda(p_1 q_1 + p_2 q_2 + \cdots p_n q_n - B)$$

and the necessary conditions of optimality are

$$0 = -\frac{\partial U}{\partial q_i} + \lambda p_i, \qquad i = 1, \cdots, n, \qquad 0 = p_1 q_1 + p_2 q_2 + \cdots p_n q_n - B.$$

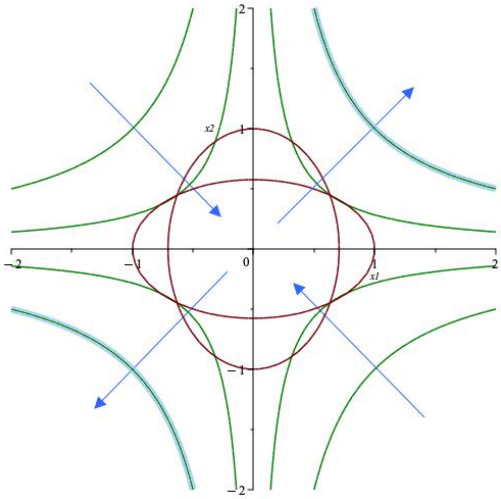
From the first n conditions, all solved for λ we obtain that the optimal multiplier $\lambda^\star$ should be such that

$$\lambda^\star = \frac{1}{p_i}\frac{\partial U}{\partial q_i}(q^\star),$$

Which states that the derivative of the optimal utility function with respect to each goods/service scaled by the cost of the goods/service is a constant. This is a well-known fact in micro-economics, which we have derived using the first order necessary conditions of optimality. [6 marks]

Answers

3. a) The admissible set is the area inside the two ellipses in the figure. The level lines are indicated in green and the arrows are in the direction of increase of the cost. The optimal solution is achieved in the second and fourth quadrant at the points in which the ellipses intersect.



A simple graphical analysis shows that all points are regular points for the constraints. Similar conclusions can be obtained by computing the Jacobian of the active constraints for all points in the admissible set and assessing their linear independence. [4 marks]

b) The Lagrangian of the problem is

$$L(x_1,x_2,\rho_1,\rho_2) = x_1x_2 + \rho_1(2x_1^2+x_2^2-1) + \rho_2(x_1^2+3x_2^2-1).$$

The first order necessary conditions of optimality are

$$0 = 4\rho_1x_1 + 2\rho_2x_1 + x_2, \qquad 0 = 2\rho_1x_2 + 6\rho_2x_2 + x_1,$$
$$2x_1^2+x_2^2-1 \le 0, \qquad x_1^2+3x_2^2-1 \le 0, \qquad \rho_1 \ge 0, \qquad \rho_2 \ge 0,$$
$$\rho_1(2x_1^2+x_2^2-1) = 0, \qquad \rho_2(x_1^2+3x_2^2-1) = 0.$$

[4 marks]

c) Using the complementarity conditions one has four options.

(O1) $\rho_1 = \rho_2 = 0$. This gives the candidate optimal solution $x_1 = x_2 = 0$, with cost $f = 0$.

(O2) $\rho_1 = x_1^2 + 3x_2^2 - 1 = 0$. This does not give any candidate solution.

(O3) $2x_1^2 + x_2^2 - 1 = \rho_2 = 0$. This does not give any candidate solution.

(O4) $2x_1^2 + x_2^2 - 1 = x_1^2 + 3x_2^2 - 1 = 0$. This gives the candidate solutions

$$x_1 = \pm\frac{\sqrt{10}}{5}, \qquad x_2 = \mp\frac{\sqrt{5}}{5}, \qquad \rho_1 = \frac{\sqrt{2}}{20}, \qquad \rho_2 = 3\frac{\sqrt{2}}{20},$$

with cost $f = -\dfrac{\sqrt{2}}{5}$.

[4 marks]

d) Note that because of symmetry of the problem the global minimisers, obtained in (O4) are not unique. The origin, obtained in (O1) is a saddle point.

[4 marks]

Answers

e) The optimal points obtained in part d) satisfy both inequality constraints with the equality sign. Hence, if one were to write the necessary conditions of optimality for the problem with equality constraints, one would obtain that the optimal points in (O4) would satisfy the necessary conditions (with Lagrange multipliers λ_1 and λ_2 equal to the optimal multipliers ρ_1 and ρ_2.

Note that, since for problem P_m there is no constraint on the sign of the multipliers, one would obtain two additional candidate points (with $\lambda_i < 0$). These would be the global maximizer of the problem.

The admissible set is composed only of four isolated points – the intersection of the ellipses. At each of these the active constraints have independent Jacobians, hence all points are regular points for the constraints. [4 marks]

Answers

4. a) The function f is a quadratic function of the form $f = \frac{1}{2}x^\top Qx + c^\top x + d$, with

$$Q = \begin{bmatrix} 8 & -1 \\ -1 & 4 \end{bmatrix}, \qquad c = \begin{bmatrix} 15 \\ 30 \end{bmatrix}, \qquad d = 0.$$

Note that Q is the Hessian matrix of f and it is positive definite (its eigenvalues are both positive). The function is therefore globally convex and radially unbounded. Its global minimiser is the stationary point $P_\star = -Q^{-1}c$, that is $P_\star = \left(-\frac{90}{31}, -\frac{255}{31}\right)$. [2 marks]

b) The iteration of the gradient algorithm from $P_0 = (0,0)$ is

$$x_{1,1} = 0 - \alpha_0\ 15, \qquad x_{2,1} = 0 - \alpha_0\ 30.$$

To compute the exact line search parameter α_0 one has to minimise the function $f(-\alpha_0\ 15, -\alpha_0\ 30)$, giving $\alpha_0^\star = 1/4$, and the new point $P_1 = \left(-\frac{15}{4}, -\frac{15}{2}\right)$. Note that $d_0 = [15,\ 30]^\top$. [4 marks]

c) To compute a direction Q conjugate to d_0 one has to find d_1, non-zero, such that $d_1^\top Q d_0 = 0$. A direct computation yields, for instance, $d_1 = [-7,\ 6]$. [4 marks]

d) To determine the point P_2 one has to consider the iteration

$$x_{1,2} = x_{1,1} + \alpha_1\ (-7), \qquad x_{2,2} = x_{2,1} + \alpha_1\ 6$$

with $x_{1,1} = -\frac{15}{4}$ and $x_{1,2} = -\frac{15}{2}$, and then minimise f with respect to α_1. This gives $\alpha_1^\star = 15/124$ and $P_2 = P_\star$. Note that this is expected, that is it is consistent with the theory, since d_0 and d_1 are Q conjugate and in both steps we have performed an exact line search. [4 marks]

e) The alternative iteration is given by

$$x_{1,k+1} = -\frac{15}{8} + \frac{x_{2,k}}{8}, \qquad x_{1,k+1} = -\frac{15}{2} + \frac{x_{1,k}}{4}.$$

Using this iteration from P_0 gives the sequence

$$x_1 = (-15/8, -15/2), \qquad x_2 = (-45/16, -255/32),$$

$$x_3 = (-2.871093750, -8.203125000),$$

$$x_4 = (-2.900390625, -8.217773438).$$

Note that $P_\star = (-2.903225806, -8.225806452)$. [4 marks]

The sequence generated by this method converges slowly to the global minimiser of f, unlike the conjugate direction method which yields convergence in two steps.

The new iteration requires minimum computation, that is two minimisations of scalar functions. [2 marks]

Answers